

SECTION 48 06 15

TURBINE OIL

PART 1 GENERAL

1.1 SUMMARY

This specification includes chlorine-free rust and oxidation inhibited (R&O) mineral oils for use in hydraulic turbine and generator bearings, Kaplan turbine hubs, hydraulic-turbine governors, and other applications, where high grade turbine oil having anti-corrosion and anti-oxidation properties is required. In addition, this specification covers delivery of 3,000 gallons of Mobil DTE Heavy (ISO VG 100) turbine oil for Bonneville Powerhouse 2.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. The year refers to the basic date of issuance of the standard or code. All Codes and Standards are to include published Interpretations and Interim Amendments or Revisions by the Standard or Code Authorities referenced must be current as of date of contract.

ASTM INTERNATIONAL (ASTM)

ASTM D92	(2018) Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D97	(2017b; R 2022) Standard Test Method for Pour Point of Petroleum Products
ASTM D130	(2019) Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test
ASTM D445	(2021; E 2022) Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D664	(2018; E 2018; E 2019) Standard Test Method for Acid Number of Petroleum Products by Potentiometric Titration
ASTM D665	(2019) Standard Test Method for Rust-Preventing Characteristics of Inhibited Mineral Oil in the Presence of Water
ASTM D892	(2018; E 2023) Standard Test Method for Foaming Characteristics of Lubricating Oils

ASTM D974	(2022) Standard Test Method for Acid and Base Number by Color-Indicator Titration
ASTM D1401	(2021) Standard Test Method for Water Separability of Petroleum Oils and Synthetic Fluids
ASTM D2270	(2010; R 2016) Standard Practice for Calculating Viscosity Index from Kinematic Viscosity at 40 and 100°C
ASTM D2272	(2022) Standard Test Method for Oxidation Stability of Steam Turbine Oils by Rotating Pressure Vessel
ASTM D3427	(2019) Standard Test Method for Air Release Properties of Petroleum Oils
ASTM D4057	(2022) Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM D4177	(2022; E 2023) Standard Practice for Automatic Sampling of Petroleum and Petroleum Products
ASTM D6304	(2020) Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration
ASTM D7042	(2021a) Standard Test Method for Dynamic Viscosity and Density of Liquids by Stabinger Viscometer (and the Calculation of Kinematic Viscosity)
ASTM D7155	(2020) Standard Practice for Evaluating Compatibility of Mixtures of Turbine Lubricating Oils
ASTM D7647	(2010: R 2018) Standard Test Method for Automatic Particle Counting of Lubricating and Hydraulic Fluids Using Dilution Techniques to Eliminate the Contribution of Water and Interfering Soft Particles by Light Extinction

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4406	(2017) Hydraulic Fluid Power - Fluids - Method for Coding the Level of Contamination by Solid Particles
ISO 11171	(2016) Hydraulic Fluid Power - Calibration of Automatic Particle Counters for Liquids

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only.

SD-04 Samples

Two 1-Gallon Samples Of Oil From Each Onsite Tank Or Truck; G

SD-07 Certificates

Certification Of Brand Name Product; G

Certification Of Clean Oil Delivery Truck and Storage Tank; G

Chain Of Custody; G

1.4 OIL DELIVERY

All oil deliveries will be held in Contractor-provided temporary onsite storage for up to 3 weeks after submitting oil samples in order to allow time for the Government to have laboratory testing performed, and review and approve results. Particular lab test results required include both property testing and compatibility testing as described below.

1.4.1 APPROVED DELIVERY TRUCK CLEANING METHODS

Truck and tanks used to deliver oil and temporarily store oil onsite must have been cleaned using a procedure which results in zero residue and therefore minimizes risk of contamination of new oil. All equipment used to deliver oil, such as pump and hoses, must also be cleaned. See subparagraphs below for approved cleaning procedures.

NOTE: If delivery tanker with onboard pump and hose is used to deliver oil to Bonneville Powerhouse 2, and is used to temporarily store that oil onsite pending laboratory test results, and the manifest/truck load documentation clearly shows that the load in the delivery truck or tank immediately prior was also Mobil DTE turbine oil (of any viscosity), then cleaning procedures in this paragraph are not required.

1.4.1.1 Oil Delivery Truck and Storage Tank Cleaning Procedure

- (a) Drain each compartment and discharge line.
- (b) Use one of the following as a cleaning fluid:
 - (i) clean unused diesel fuel with less than 0.1% biodiesel and less than 0.2% total additives
 - (ii) kerosene
 - (iii) gasoline
 - (iv) jet fuel
- (c) For tanks or single-compartment trucks, utilize a minimum of 50 gallons of cleaning fluid in the compartment. For triple-compartment trucks, utilize a minimum of 10 gallons of cleaning fluid in each compartment.
- (d) Introduce cleaning fluid from the top of each compartment, such as by attaching a spinner ball or spray head to man-way on top of compartment. Attach the discharge hose to the discharge port of the compartment. Circulate the cleaning fluid through the discharge hose

to the spinner ball or spray head. Repeat this for a minimum of 30 minutes. If there are multiple discharge ports, clean each discharge port (i.e., circulate the cleaning fluid through each discharge port during the cleaning process).

- (e) After cleaning process is complete, discharge the cleaning fluid through the discharge hose to a proper container.
- (f) After cleaning, check the compartment for residual product, emulsion, dirt, strong odor, or foreign debris. If compartment does not look clean, repeat flushing procedure.
- (g) After cleaning, drain all discharge ports on the transport, including all manifold points of discharge and all internal valves opened, and check for residual product or cleaning fluid. Slight elevation of the transport front will facilitate drainage. Total residual cleaning fluid remaining in the entire transport or tank must not be more than 5 gallons.
- (h) Use a nitrogen or dry compressed-air blowdown.

1.4.1.2 Onboard Hose Cleaning

- (a) Drain and flush all hoses used to deliver product with cleaning fluid. Inspect, and repeat as needed until clean.
- (b) Drain cleaning fluid from hoses and blow dry with nitrogen or air.

1.4.1.3 Onboard Pump Cleaning

- (a) Clean pump to be visibly free of dirt, water, or other contamination.
- (b) Pump a minimum of 4 gallons of Mobil DTE Heavy turbine oil.

1.4.2 DELIVERY HOSE REQUIREMENTS

Hoses must have a rated pressure of no less than 125 psi or 150% of the relief valve setting on the delivery pump system, whichever is greater. Delivery hose must be not less than 2 inches in nominal diameter. Hoses must be cleaned by a rag or pig to remove any loose visible contamination including dirt, sludge, varnish, lint, metal or any other visually observable debris. Before connecting the hoses allow the Government to visually check to confirm that the cleaning has occurred.

1.4.3 OIL SAMPLING

Obtain all samples per [ASTM D4057](#) or [ASTM D4177](#).

PART 2 PRODUCTS

2.1 CERTIFICATION OF BRAND NAME PRODUCT

Submit certification such as a letter with the manufacturer's certificate

of analysis, Product Data Sheet or Technical Data Sheet (PDS/TDS), and Material Safety Data Sheet (MSDS) attachments that the product to be delivered per these specifications is the exact brand name specified, Mobil DTE Heavy (ISO VG 100) turbine oil. No "equivalent" products are acceptable for this requirement.

2.2 GENERAL REQUIREMENTS

2.2.1 Properties

Provide turbine oil which is a blend of virgin petroleum-based stocks plus additives, free of chlorine, resulting in high-grade turbine oil having anti-rust and anti-oxidation properties suitable for use in hydraulic turbines, generator bearings, Kaplan turbine hubs and related applications.

2.2.2 Compatibility

Delivered oil must be Visually Compatible with in-service oil per Tier 1 of [ASTM D7155](#).

2.2.3 Chemical And Physical Characteristics

Chemical and physical characteristics of oil must meet or exceed the requirements listed in TABLE 1 when tested according to the standards indicated.

TABLE 1: CHEMICAL AND PHYSICAL CHARACTERISTICS REQUIREMENTS AND TEST METHODS FOR RUST AND OXIDATION (R&O) INHIBITED ISO 100 TURBINE OILS		
Chemical and Physical Characteristics	Requirements ISO 100 Oil	Test Method
Viscosity at 40 C, centistokes (cSt)	95-102	ASTM D445 or ASTM D7042
Viscosity Index, minimum	95	ASTM D2270
Flash Point, minimum, C (F)	210 (410)	ASTM D92
Pour Point, maximum, C (F)	-9 (16)	ASTM D97
Acid Number (AN) mg KOH/g, maximum	0.15	ASTM D664 or ASTM D974
Oxidation Stability by Rotating Pressure Vessel Oxidation Test (RPVOT), minutes, minimum	500	ASTM D2272
Rust Preventive Characteristics, Procedures "A" and "B"	Pass	ASTM D665
Water Content, parts per million (ppm), max	For Information Only	ASTM D6304
Water Separability of Petroleum Oil	40-40-0 (30)	ASTM D1401
Corrosion from Oil by Copper Strip Tarnish Test	Class 1	ASTM D130
Foaming characteristics after 5 minutes blowing period:		

TABLE 1: CHEMICAL AND PHYSICAL CHARACTERISTICS REQUIREMENTS AND TEST METHODS FOR RUST AND OXIDATION (R&O) INHIBITED ISO 100 TURBINE OILS		
Chemical and Physical Characteristics	Requirements ISO 100 Oil	Test Method
Sequence 1, foam volume in ml, maximum	100	ASTM D892 (Option "A" excluded) NOTE: These numbers are the foam "tendency" results.
Sequence 2, foam volume in ml, maximum	50	
Sequence 3, foam volume in ml, maximum	100	
Foaming characteristics after 10 minutes settling period:		
Sequence 1, foam volume in ml, maximum	10	ASTM D892 (Option "A" excluded) NOTE: These numbers are the foam "stability" results.
Sequence 2, foam volume in ml, maximum	0	
Sequence 3, foam volume in ml, maximum	10	
Air Release Properties, minutes, max.	30	ASTM D3427
Cleanliness, Reported per ISO 4406 as Iso Code Particle Count, for particles greater than 4, 6, and 14 m (c)	For Information Only	ASTM D7647 (ISO 11171 Cal)
Appearance	Clear and Bright	Visual Observation

2.3 DEGRADATION

The physical and chemical properties of the oil, with the exception of foam tendency, must not be degraded (changed from the specified values) by filtration through two-micron mechanical type filters, by centrifugal purification, two-micron coalescing filters, balanced charge agglomeration or by vacuum type purifier, all of which have been designed for turbine oil.

2.4 HOMOGENEITY

Additive agents must remain uniformly distributed throughout the oil at all temperatures above the pour point and up to 250 degrees F. When the oil is cooled below the pour point, it must regain homogeneity while standing at temperatures of 10 degrees F above the pour point, and retain clear and bright appearance.

PART 3 EXECUTION

3.1 DELIVERY OF NEW OIL

Deliver new oil in accordance with specifications and in coordination with the Contracting Officer for the delivery schedule. Spill prevention must be set up and observed by the Government prior to delivering any oil.

3.1.1 Chain of Custody

Upon delivery of oil, provide a chain of custody report that includes documentation such as the bill of lading. Reports must track the oil, referenced by its batch number, from its origin to final delivery signed and dated by a designated representative.

3.1.2 Certification of Clean Oil Delivery Truck and Storage Tank

Prior to or upon delivery of oil, submit certification that the truck used for that particular delivery was cleaned per an approved procedure, or documentation that clearly shows the load immediately prior was also Mobil DTE turbine oil (of any viscosity). Upon onsite storage of oil, submit certification that storage tank was cleaned per an approved procedure, if applicable.

3.1.3 Temporary Onsite Oil Storage

Provide means of temporary new oil storage to hold entire delivery of new oil or leave oil truck tanker on site as new oil storage while laboratory testing is performed. Oil containment measures must be provided with any form of temporary storage except in the instance of a double-walled tank. Coordinate with Project personnel for staging locations.

Upon Government approval of laboratory test results, Contractor must pump approved oil into the Government's storage tank.

3.2 Two 1-Gallon Samples Of Oil From Each Onsite Tank Or Truck

At the point of oil delivery, alert the Project POC to witness oil sampling from truck/tank onsite. Obtain two 1-gallon samples in a manner specified in [ASTM D4057](#) and provide them to the Government.

3.3 INSPECTION AND ACCEPTANCE

The Government may perform such tests as are deemed necessary to determine whether the oil meets ASTM D7155 compatibility requirements as well as the specifications values listed in TABLE 1. The delivered oil must remain onsite in delivery truck or Contractor-provided storage tank or tote, and cannot be used until the test results are received from the laboratory. Should the oil fail to meet requirements, or show results incompatible with being the specified name brand required, or be so contaminated at the delivery point that it fails compatibility testing with in-service oil, the Contractor is responsible for removing oil and disposing of the rejected oil and replacing the oil at no additional cost to the Government, within 3 months of initial delivery.

-- End of Section --